802.11g

After further research, in mid 2003, the 802.11g protocol was released. This protocol was fully backward-compatible with 802.11b, mainly because it used the same 2.4 GHz RF spectrum, and also offers 54 Mbps theoretical throughput, like the 802.11a standard. The drawback of this backward-compatibility is the fact that an entire network can be slowed down to a fraction of 802.11g's supported throughput speed, just by introducing an older 802.11b device into it. 802.11g offers lower ranges than 802.11b, but makes up for this in terms of actual data throughput limits over short distances.

An 802.11g PCI wireless card

802.11n

This standard was approved for research in early 2004, and results are expected somewhere in 2006. The 802.11n standard will offer a throughput of 100 Mbps, and will have a higher range than the previous standards. Task group 16 of IEEE 802 is working on something even more exciting, though.

802.16

Popularly called WiMAX (Worldwide Interoperability for Microwave Access), this standard is developed by working group 16 of the wireless committee. It is a standard that will complement current Wi-Fi setups and is solely a MAN technology. It can con-